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Hybrid modeling of cross-flow filtration: Predicting the flux evolution and duration of ultrafiltration processes

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ABSTRACT

Cross-flow ultrafiltration is a powerful tool used in bioprocesses to concentrate and separate biopharmaceuticals. However, there is a major challenge for its effective use: the fouling rate and decrease in the permeate flux vary substantially depending on the concentration and characteristics of the incoming feed, which causes variations in the process durations. We developed a hybrid model for use in predicting the flux evolution and duration of cross-flow ultrafiltration processes for various proteins, membrane types, input parameters, and filtration modes. The trained hybrid model is able to determine the process duration with an average normalized root-mean-square error of less than 6.2. It has superior adaptation to varying filtration characteristics compared to the mechanistic film theory model and can handle both batch- and fed-batch ultrafiltration using the same training set. In addition, the presented hybrid model can be used as a digital twin to simulate virtual processes with varying input parameters and different batch modes, which is a valuable tool for efficient process development or optimization.

1. Introduction

Cross-flow ultrafiltration (UF) is often employed in downstream processes of therapeutic biomolecules to concentrate a product based on its size. It is an attractive workhorse for research, development, and manufacturing purposes, since it can be scaled up by engineering-based approaches and can handle large volumes.

A challenge of UF is that the operation time can vary strongly depending on the characteristics of the bulk solution regarding, e.g., protein attributes [1], content of nucleic acids [2], or aggregates [3]. Over time, soluble macromolecules accumulate on the filter surface and form a concentration polarization (CP) layer and/or a gel layer [4] that decreases the permeate flux (hereafter referred to as flux). This accumulation is a complex mechanism that depends on the concentration and species in the bulk solution in combination with process conditions that may lead to adsorption [5], different pore-blocking mechanisms [6], and other interactions with the membrane [7]. This complexity makes it difficult to predict when the filtration process will be completed. A great deal of effort has been devoted to describing mechanistically these phenomena and their influence on the flux [8,9]. However, phenomena that are not fully understood cannot be represented in mechanistic models. An alternative is data-driven models that can help describe such phenomena without underlying physical explanations, although their disadvantage is that they provide little insight into the functioning of the system.

Hybrid modeling can combine the benefits of both modeling approaches. A hybrid model consists of a data-driven, nonparametric part that learns from data and a mechanistically described, parametric part. These parts are hereafter referred to as black box and a white box. The black box represents the correlation of input variables with output variables using the weights/coefficients of a nonparametric function, e.g., an artificial neural network (ANN), partial least squares regression (PLS), structured additive regression, or random forest, among others. The white box represents a mechanistically or phenomenologically well-understood function that can be derived from conservation, kinetic, or thermodynamic laws. In the black box model, the structure of the functions and their parameters are derived from data. In a hybrid model, the black and white boxes can be combined in multiple ways, basically as serial or parallel hybrid models. In a serial hybrid model, the output from the black box is used as input for the white box or vice

Abbreviations: ANN, artificial neural network; BSA, bovine serum albumin; c_B, bulk concentration; CF, cross-flow velocity; CP, concentration polarization; MWCO, molecular weight cutoff; NRMSE, normalized root-mean-squared error; PESU, polyether sulfone; RMSE, root-mean-squared error; SEC, size exclusion chromatography; TMP, transmembrane pressure; UF, ultrafiltration

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Nomenclature

List of symbols

A	membrane area [m^2]
c_B	bulk concentration [g/L]
c_{fb}	concentration in fed-batch bottle [g/L]
c_G	gel layer concentration [g/L]
dt	time increment [s]
J	permeate flux [LMH] or [$m^3/m^2{\cdot}s$]
k	mass transfer coefficient [LMH]
p_f	feed pressure [bar]
p_p	permeate pressure [bar]
p_r	retentate pressure [bar]
R_{obs}	observed retention coefficient [-]
V_B	bulk/reservoir volume [mL]
V_{fb}	fed-batch bottle volume [mL]
V_p	permeate volume [mL]

versa. This option is used if a part of the model is poorly understood. For filtration, this might be the impact of feed composition on the flux. Parallel hybrid models are used when mechanistic models are available, but the predictions need to be improved with a black box model [10]. In some cases, a combination of both forms is required [11].

Hybrid models are widely used to describe bioprocesses [12,13]. Several papers have described hybrid models used for filtration applications, mostly in the field of wastewater treatment. They have been successfully used to predict the flux dynamics and different cleaning strategies for UF-based wastewater clarification by combining Darcy's law with an ANN [14]. A similar model was employed at industrial scale to predict the fouling probability for dead-end UF applied to wastewater treatment [15]. Another publication showed that hybrid models can be used to predict counterion fluxes across membranes in membrane-supported biofilm reactors [16]. Additionally, it has been shown that ANNs are able to make an in-process flux prediction within the same UF run, but not for completely new UF runs with varying input parameters [17].

In this study, we present a hybrid model that can predict multiple steps ahead. It describes the future change in flux over time and thereby determine the duration of a UF process with the varying input parameters transmembrane pressure (TMP), cross-flow velocity (CF), and protein bulk concentration (c_B). This novel hybrid model structure was tested on two model proteins—bovine serum albumin (BSA) and lysozyme—using two different membrane materials. The resulting hybrid model was compared to the film theory model. Further, the hybrid model was tested in batch and fed-batch continuous UF modes. In the latter case, additional unconcentrated feed volume was continuously added to the reservoir. Having a single multi step ahead model structure with the ability to describe ultrafiltration processes for different modes, proteins and membrane types is a new approach that allows the implementation of a digital twin and model predictive control [18].

2. Experimental

2.1. Equipment and chemicals

The UF runs were performed on an ÄKTACrossflow system (GE Healthcare Life Sciences, Uppsala, Sweden) controlled by UNICORN 5.31 software. The reservoir size for the feed solution was 1100 mL. Two types of Sartocon Slice cassettes (Sartorius AG, Göttingen, Germany) were used: a hydrophobic polyether sulfone membrane (PESU) and a hydrophilic, stabilized cellulose-based membrane (Hydrosart). For both cases, the filter area was 200 cm^2. Two model proteins were used: BSA and lysozyme from chicken egg white (both Sigma-Aldrich, St. Louis, MO, USA). The experiment used 50 mM PBS with pH 8 as a filtration buffer. A summary of the performed protein-membrane combinations and the training and test set sizes are provided in Table 1. When choosing pore sizes of the membranes we considered the recommendation of the supplier and selected membranes with a molecular weight cutoff (MWCO) that is 3–5 times less than the molecular weight of the target protein. For BSA with a molecular weight of ~66 kDa a 10 kDa MWCO and for lysozyme (14.7 kDa) a 5 kDa membrane was used. This rule was applied for all cases except for the combination lysozyme-PESU, where a 5 kDa MWCO membrane led to strong fouling and complete filter clogging. In that case, a 10 kDa MWCO membrane was used instead.

Before the experiments were performed, the filters were flushed with at least 1 L deionized water and conditioned with the filtration buffer until a stable conductivity signal was reached. After the experiment, the filters were flushed with 1 L of deionized water, cleaned for at least 60 min with a clean-in-place (CIP) solution, and again flushed with 1 L of deionized water. The CIP solution for the PESU membranes was 0.4 M NaOH with 300 ppm NaOCl and, for the Hydrosart membranes, 1 M NaOH in water. Before and after each experiment, the clean water flux was measured to ensure the same starting conditions for each run.

2.2. Training and test data generation

The data for training the black box model were collected using the UNICORN built-in process optimization tool. The schematic depiction of the cross-flow filtration unit is shown in Fig. 1. For a given c_B, first the CF was adjusted (employing the feed pump) to the lowest flow rate, followed by stepwise TMP variations from lowest to highest value employing the valve on the retentate-side P_R (Fig. 1c). Afterward, the CF was increased stepwise. Throughout this scouting, the permeate was redirected into the reservoir to keep the c_B constant (Fig. 1a). For each c_B, three CFs and five TMPs were tested, and the resulting fluxes were recorded. To increase the c_B, the sample was concentrated (Fig. 1b) until the desired bulk reservoir volume was reached (Fig. 1d), and then the next scouting run was performed (Fig. 1a and c). After each concentration step, a sample was taken from both the retentate and permeate and measured using SEC-HPLC.

After the data collection for the black box model training, UF test runs with varying initial c_B, CF, and TMP were performed. The decrease in flux over the process time was compared to the predictions from the hybrid model. For each UF run, a 1000 mL protein solution was prepared with a defined protein concentration that was concentrated at constant TMP and CF to a retentate volume of approximately 40 mL. When the UF was performed in fed-batch mode, a fresh sample was added to the bulk reservoir according to the permeate flow to keep the reservoir level constant.

Table 1
Summary of the combinations of model proteins and membrane types with training set size and number of test runs. The combination BSA and Hydrosart was tested with five runs in batch mode and three runs in fed-batch mode. All other combinations were performed in batch mode.

Protein	Material	MWCO [kDa]	Size of training set	scouted c_Bs	c_B range of training set [g/L]	Number of test runs
BSA	Hydrosart	10	365	25	1.6–230.6	5 + 3
BSA	PESU	10	173	13	1.7–301.8	5
Lysozyme	Hydrosart	5	190	13	1.6–226.4	5
Lysozyme	PESU	10	193	13	1.7–235.3	2

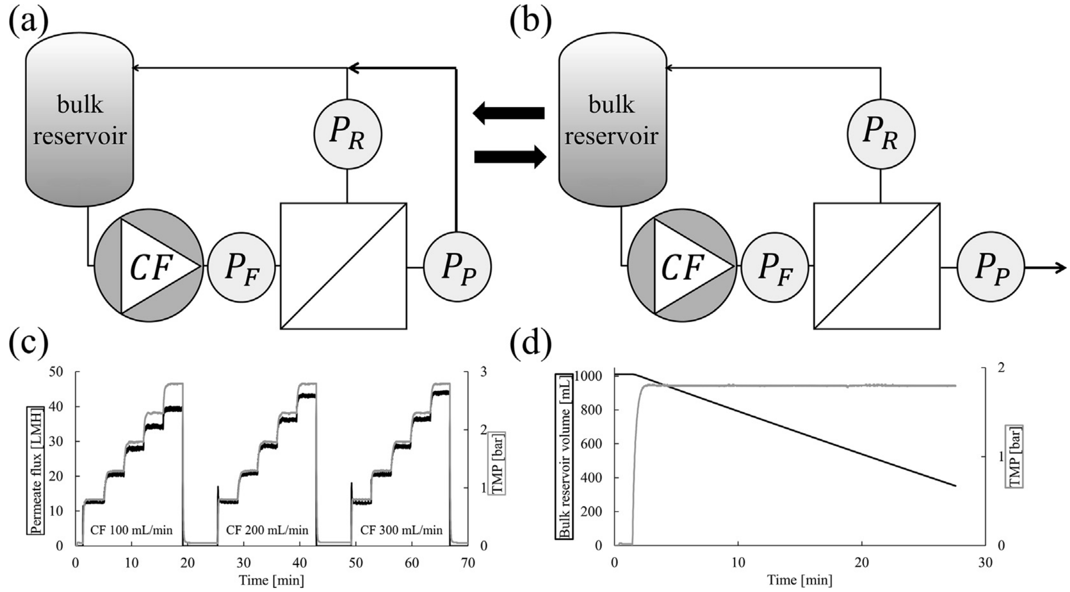

Fig. 1. Schematic overview of the cross-flow filtration setup. (a,c) During training data generation, TMP and CF scouting the permeate was redirected into the bulk reservoir keeping c_B constant. (b,d) To increase c_B during the training data generation the sample was concentrated and the bulk reservoir volume was reduced. The same setup was used to generate the test sets. For the training data generation (a) and (b) were used in an alternating mode.

2.3. Protein analysis

BSA and lysozyme concentrations were determined by analytical size exclusion chromatography (SEC) on an Agilent 1290 LC system (Santa Clara, California, USA) using a TSKgel G3000SWXL column (5 µm, 7.8 × 300 mm; TOSOH, Shiba, Tokyo, Japan). The separation was performed under isocratic conditions with 50 mM sodium phosphate and 200 mM NaCl (pH 6.5) as the running buffer at a flow rate of 0.4 mL/min. To estimate the protein concentration during training and testing standard calibrations for lysozyme and BSA were prepared in 50 mM PBS, pH 8 filtered through a 0.22 µm Millex-GV filter (Merck Millipore, USA). For lysozyme and BSA four standards between 0.1 and 1.0 g/L and 0.1 g and 1.5 g/L were used, respectively (see calibrations and chromatograms in Fig. A.4 and A.5). The samples from the training and test runs were diluted to fit the calibration range of each protein. The injection volume was 10 µL per analysis.

2.4. Hybrid modeling

2.4.1. Black box model

The aim for the black box part of the hybrid model was to predict the flux (output parameter) for a combination of three input parameters: CF, TMP, and c_B. In this model, an ANN was used as the black box. Computations were performed using MATLAB R2018b. The ANN

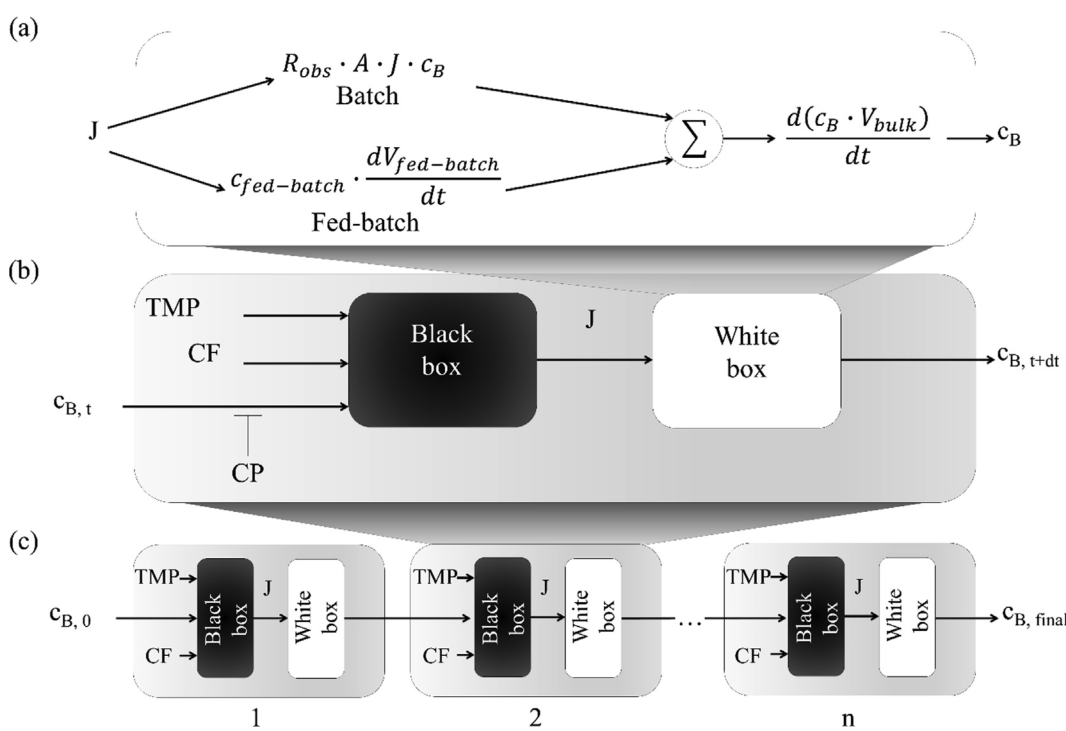

Fig. 2. Schematic representation of the (a) white box, (b) hybrid model, and (c) its multistep-ahead prediction capability.

Table 2
Summary of the test sets for different combinations of model protein and membrane type with varying initial c_B, CF, and TMP and resulting NRMSE.

Protein and membrane	Initial c_B [g/L]	CF [mL/min]	TMP [bar]	NRMSE [%]
BSA Hydrosart 10 kDa	2.5	100	1.8	6.6
	7.5	200	1.8	2.4
	23.2	300	1.8	3.1
	14.2	150	1.3	2.9
	4.1	250	2.5	6.1
BSA PESU 10 kDa	7.9	200	1.8	9.1
	5.5	250	2.5	9.2
	11.2	230	1.8	8.3
	6.5	250	1.5	11.0
	12	300	1.8	14.0
Lysozyme Hydrosart 5 kDa	5.1	200	1.8	2.8
	5.4	200	2.3	5.8
	4.3	300	1.8	3.7
	9.3	250	2.5	4.4
	1.9	250	2.1	5.3
Lysozyme PESU 10 kDa	5.0	200	1.8	7.5
	4.7	300	1.8	8.3
BSA Hydrosart 10 kDa Fed-batch	12.0	260	2.4	2.6
	4.2	200	1.8	5.9
	7.9	230	2.0	4.5

requires optimization of two tuning parameters—the number of hidden neurons and the weight decay—to avoid under- and overfitting. The model was tuned by varying the number of neurons from 1 to 10 in one hidden layer. The ANN was set up with the *feedforwardnet* function and trained with the *trainbr* function. This latter function uses Bayesian regularization backpropagation to avoid overfitting by automatically penalizing large weights (weight decay). The best setup contained one hidden layer with five neurons (Fig. A.1), a sigmoid activation function in the hidden layer, and linear activation functions in both the input and output layers. The inputs and output were preprocessed using normalization between 0 and 1.

2.4.2. White box model

The applied white box predicted c_B and the incremental permeate volume increase (dV_p) after a given time interval (dt) using the predicted flux (J) output of the black box model and the filter area (A), shown in Eq. (1). The observed retention coefficient R_{obs} is a protein-membrane specific value between 0 and 1. For all protein membrane combinations used in this work, R_{obs} was 1.

$$\frac{dV_p}{dt} = A \cdot J \tag{1}$$

$$\frac{dV_B}{dt} = -A \cdot J + \frac{dV_{fb}}{dt} \tag{2}$$

$$\frac{d(c_B \cdot V_B)}{dt} = R_{obs} \cdot A \cdot J \cdot c_B + c_{fb}\frac{dV_{fb}}{dt} \tag{3}$$

The white box (Fig. 2a) consists of Eq. (1), Eq. (2), and Eq. (3) employed for both batch and fed-batch filtration. Eq. (2) was used to calculate the reservoir bulk volume (V_B) and Eq. (3) the c_B for the next time interval.

In the case of a batch UF, $dV_{fb}/dt = 0$. Since $R_{obs} = 1$, $d(c_B \cdot V_{fb})/dt = 0$ as well.

For fed-batch processes, Eq. (2) assumes an immediate refilling of the reservoir with a fresh unconcentrated sample (V_{fb}, c_{fb}) as soon as the volume passes through the membrane on the permeate side; therefore, V_B remains constant with $dV_B/dt = 0$. Eq. (3) calculates c_B for the next time interval. The constant reservoir level was achieved by employing a level sensor that controlled the refilling rate of the reservoir. After the fed-batch bottle was emptied, the white box continued predicting the future c_B assuming $dV_{fb}/dt = 0$.

2.4.3. Training and test data

In the training data set for each c_B, three CFs (100, 200, and 300 mL/min) and five TMPs (0.8, 1.3, 1.8, 2.3, and 2.8 bar) were tested, and the resulting fluxes were recorded. The amount of c_Bs for each protein-membrane combination is given in Table 1 including the c_B ranges. For each protein-membrane combination two training runs were performed, leading to a total of eight training experiments. A detailed description of all scouted c_Bs is given in Table A.1 and all training observation are summarized in Table A.3-A.10.

After the data collection for black box model training, UF test runs with varying initial c_B, CF, and TMP were performed. The decrease in observed flux over the process time was compared to the predictions from the hybrid model.

To assess the predictive power of the hybrid model, the root-mean-square error (RMSE) and normalized RMSE (NRMSE) were calculated in Eq. (4) and Eq. (5), respectively, by comparing the predicted fluxes at each time point to the measured fluxes:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4}$$

$$NRMSE = 100 \cdot \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}}{y_{max} - y_{min}} \tag{5}$$

where n is the number of target fluxes y_i and the corresponding predicted fluxes $\hat{y}_i$. The normalization y_{max}-y_{min} enables a fair comparison of different protein- and membrane-specific attributes, e.g., protein and pore size. Table 2 gives the combinations of inputs for all test experiments.

2.4.4. Multistep-ahead hybrid model

A schematic overview of the hybrid model is given in Fig. 2b. It used the three inputs TMP, CF, and corrected c_B to estimate the flux via the black box. In a second step, the white box calculated the future c_B. Before feeding c_B to the black box, it was corrected by taking the CP layer characteristics into account (Section 3.2).

The developed hybrid model was capable of predicting multiple steps ahead, as depicted in Fig. 2c. Here, the c_B output from the first hybrid model was used to calculate future fluxes and c_Bs until a desired stop criterion was reached—in our case, the final retentate volume.

The established hybrid model was used to predict the flux evolution and UF duration for various combinations of TMP, CF, and c_B. As an example, for a desired process duration, CF, and c_B, the required TMP was predicted using an iterative solver, which gradually approached the optimal TMP by iteratively changing TMP values. The TMP optimization was performed using Eq. (6). When the resulting endpoint was larger than $TMP_{desired}$, TMP_{min} was replaced with the current TMP, and if it was smaller than $TMP_{desired}$, the current TMP replaced TMP_{max}. If the error between the predicted and the desired duration became lower than the average error of the test set, the iteration stopped. A boundary condition for the predictive model was the feed pressure, which is a limitation of the filter and the filtration device. In this case, the maximum feed pressure and TMP was 4 bar. This approach can be used for estimating CF and c_B as well.

$$TMP_{desired} = \frac{TMP_{max} + TMP_{min}}{2} \tag{6}$$

2.5. Film theory model

The suggested hybrid model was compared to the well-established mechanistic film theory model. Film theory derives from the mass-transfer model described by convective transport toward the membrane and back-diffusion caused by the concentration gradient [19]. According to the film theory model, the flux J is related to c_B by

$$J = k \cdot ln\left(\frac{c_G}{c_B}\right) \tag{7}$$

where c_G is the gel-layer concentration at the membrane surface and k is the mass transfer coefficient that depends on the diffusion coefficient and the thickness of the gel layer [19]. The film theory model is valid in the pressure-independent region. The parameters k and c_G are extrapolated for constant TMP and CF from the slope and the intersection with the abscissa in the J-ln(c_B) plot. To compare the hybrid model with the film theory model, the black box was replaced by the film theory model in Eq. (7) using the parameters k and c_G instead of TMP and CF.

3. Results and discussion

3.1. Discussion of experimental data

Data were generated for four combinations of protein and membrane types (Table 1). Five TMPs (0.8, 1.3, 1.8, 2.3, and 2.8 bar), three CFs (100, 200, and 300 mL/min), and protein concentrations c_B from 1.6 g/L to 301.8 g/L were measured, leading to a total of 173–365 observations for each training set (see Table 1). In general, the flux increased with CF (Fig. A.3) and TMP (Fig. 3). With increasing c_B, the flux differences between each TMP interval became smaller since the system approached the pressure-independent region. In this region, the mass transfer equilibrium becomes independent from TMP, and the flux stays constant. A detailed explanation of the flux behavior is given in Section 3.4. The blank samples (c_B = 0 g/L) were excluded from the training set, since they led to overestimation at low c_B values due to the high deviation between the blank and the protein samples.

3.2. Concentration polarization (CP) correction

The SEC results indicated that all permeate solutions contained less than 0.05 g/L of protein. Therefore, the retention coefficient R_{obs} = 1 was assumed for calculating the c_B for the training and test set. Fig. 4b shows that the measured c_B using the Hydrosart membrane deviated from the calculated values at higher c_B.

We observed that Hydrosart membranes exhibited a more pronounced CP layer compared to PESU membranes. Also, the protein concentration in the CP layer on the Hydrosart membrane increased over time, which caused the deviation between the measured c_B and the calculated c_B. For the PESU membrane the measured and calculated c_B were identical, indicating a very small to non-existing CP layer on top of the membrane surface at any c_B. PESU membranes are more prone to fouling, and a thin fouling layer is formed, e.g., by adsorption [20]. A quadratic polynomial function was used to derive the measured c_B from

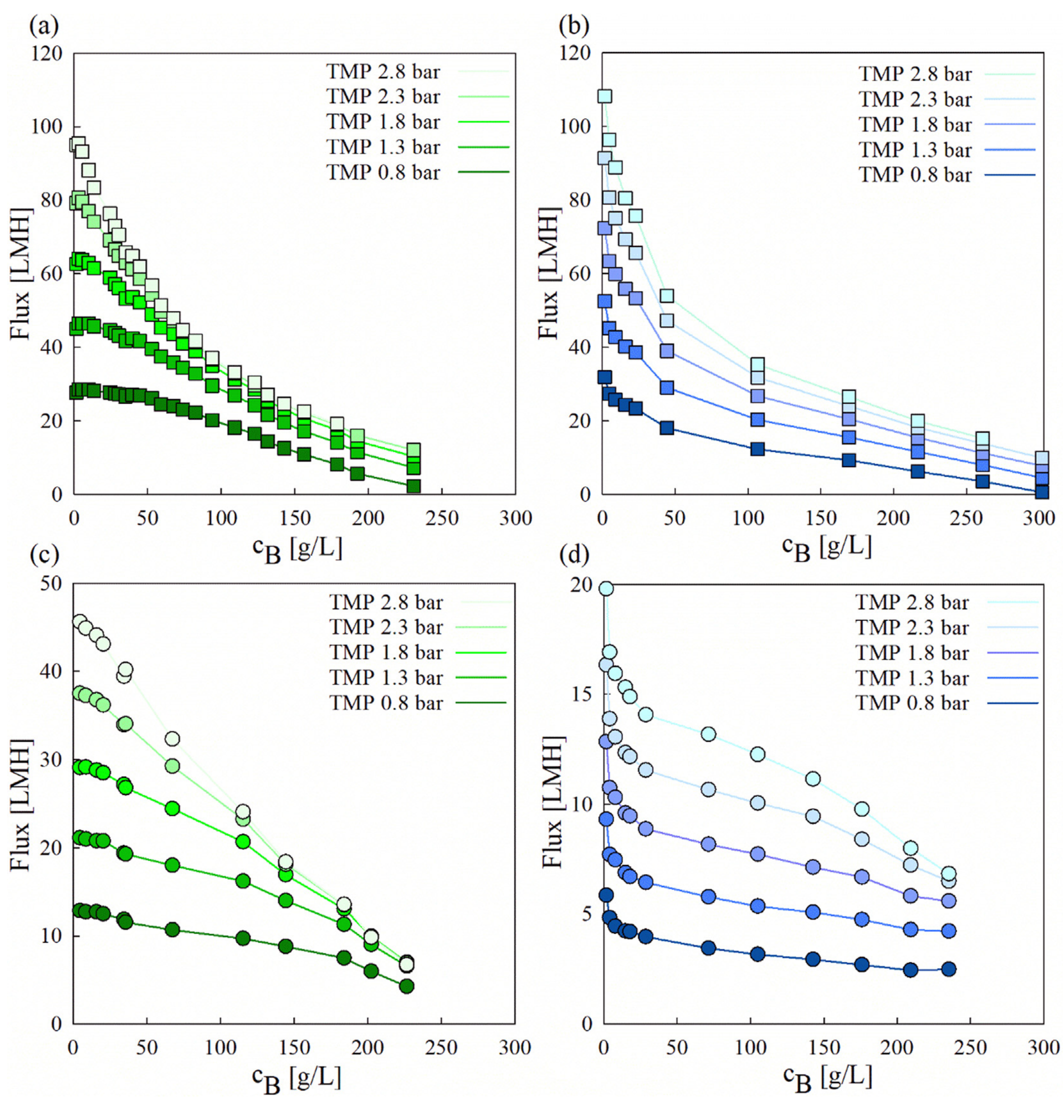

Fig. 3. Flux vs. c_B at CF 200 mL/min of BSA (squares) and lysozyme (circles) with Hydrosart (green) and PESU (blue) membrane. (a) BSA with 10 kDa Hydrosart, (b) BSA with 10 kDa PESU, (c) lysozyme with 5 kDa Hydrosart, and (d) lysozyme with 10 kDa PESU.

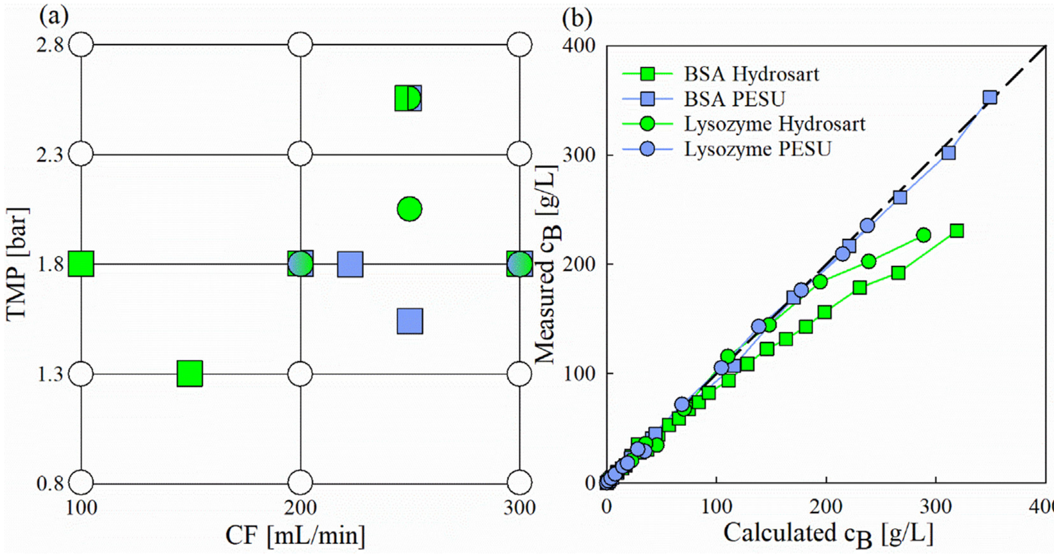

Fig. 4. (a) Schematic depiction of TMP and CF chosen for UF runs of the test set. The white dots indicate the parameter values of the training set. The colored symbols indicate the input parameters of the test sets for the combinations BSA-Hydrosart (green squares), BSA-PESU (blue squares), lysozyme-Hydrosart (green circles), and lysozyme-PESU (blue circles). (b) Deviations between calculated and measured c_Bs for all combinations of proteins and membranes of the training sets, including the identity line (dotted line). The concentration steps was performed with TMP 1.8 bar and CF 200 mL/min.

the calculated c_B, and this function was calculated using the values from Fig. 4b for each protein-membrane combination.

3.3. Model performance assessment

First, the hybrid model was tested on the training data using 5-fold cross validation (CV). The results for all protein-membrane combinations are given in Table 3. The RMSE correlates with the highest absolute fluxes (Fig. 3), with lysozyme-PESU exhibiting the lowest and BSA-PESU the highest error. The NRMSE accounts for the different flux maxima and minima in the training sets and is better suited for comparing the four protein-membrane combinations. The ANN shows the lowest error for BSA-Hydrosart followed by the other three combinations, lysozyme-Hydrosart, BSA-PESU, and lysozyme-PESU. BSA-Hydrosart shows the smoothest transition between the training fluxes (Fig. 3a and A.2a) with no irregularities at the lower concentrations as determined for the other conditions. Additionally, the training set size, at 395 observations, was twice as large as the other training sets.

The optimal ANN structure was determined by varying the number of hidden nodes from 1 to 10 and tested on the training set using 5-fold CV. The ANN with five hidden nodes was chosen for all protein-membrane combinations, because it exhibited a low RMSE and its error did not reduce further as more nodes were added (Fig. A.1). The hybrid models were tested on data that were generated independently from the training set. For an entire test set of each protein-membrane combination, 2-5 batch UF runs were performed with varying initial c_B, TMP, and CF within the training space. Fig. 4a shows all TMP and CF combinations that were included in the test sets and Table 2 contains a list of all input parameters. All TMP and CF combinations of the test set were within the ranges used for training set generation (white dots). Most UF runs were performed in the upper right corner, since high TMP and CF values are the most relevant factors for reaching high fluxes and shorter operation times.

3.4. Filtration duration

The established hybrid model was tested on UF processes under constant CF and TMP conditions that were within the range of the training region. In each iteration of the hybrid model, the updated c_B was corrected using a second-order polynomial based on the data in Fig. 4b. The flux evolution throughout the UF runs are depicted in Fig. 5 for all four protein-membrane combinations.

The fluxes of BSA filtration using Hydrosart and PESU membranes were similar. However, experiments using the PESU membrane showed a higher initial flux peak. After approximately 10 min, the flux reached a pseudo-steady state, similar to the experiments with the Hydrosart membrane. At the chosen buffer pH of 8 both BSA (pI 4.7) and the PESU membrane (pI $<$ 4.5 [21]) are negatively charged, which leads to repulsion and therefore higher fluxes at the beginning of the UF [21,22]. Under these conditions the fouling of the PESU membrane is negligible [23] not leading to stronger flux reduction compared to the Hydrosart membrane. The initial high flux observed for the concentration of BSA by PESU membranes was not modeled correctly. Also, the BSA-PESU training set (Fig. 3b) exhibited higher fluxes at low c_B compared to those for BSA with Hydrosart (Fig. 3a). However, for higher c_B, the Hydrosart membrane showed higher fluxes since it was less prone to fouling. Similar results were already described for microfiltration [24]. Lysozyme-Hydrosart experiments showed lower fluxes compared to BSA, because the smaller pores of the membrane used in this case caused higher resistance of this membrane toward the applied pressure. However, the flux evolution was similar to BSA-Hydrosart experiments. The concentration of lysozyme by PESU membranes exhibited much lower fluxes compared to the case when Hydrosart was used, and the initial flux peak was much smaller compared to the BSA-PESU setup. Both phenomena are due to the strong attraction between the positively charged lysozyme (pI 11 [25]) and negatively charged PESU membrane (pI $<$ 4.5 [21,26]) at pH 8 and

Table 3
RMSE and NRMSE of 5-fold CV for the four training sets.

Error measure	BSA Hydrosart 10 kDa	Lysozyme Hydrosart 5 kDa	BSA PESU 10 kDa	Lysozyme PESU 10 kDa
RMSE [LMH]	1.02	1.73	3.40	0.59
NRMSE [%]	1.61	3.45	4.46	2.90

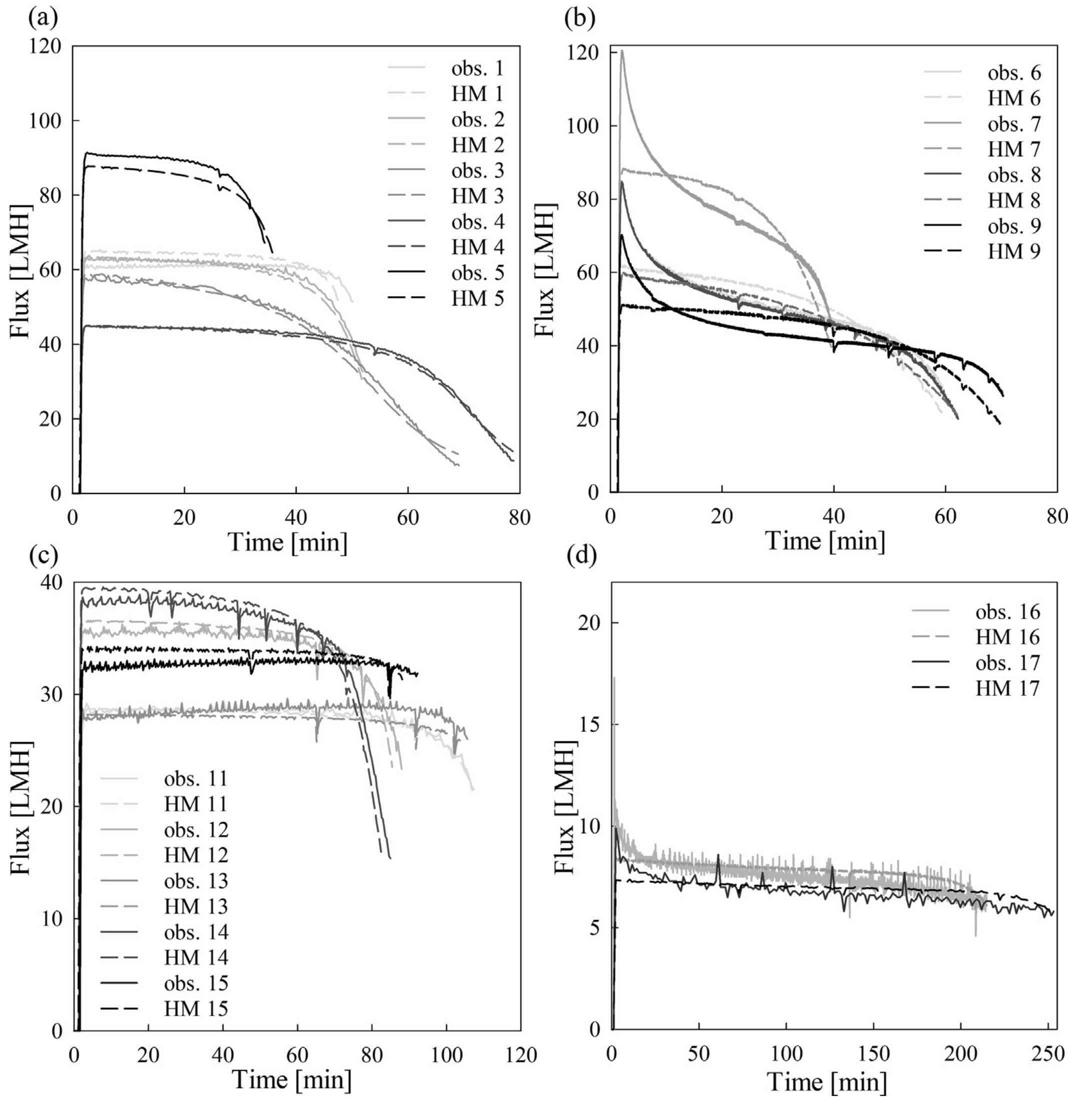

Fig. 5. Summary of test sets with varying initial c_B, TMP, and CF for the protein-membrane combinations (a) BSA-Hydrosart 10 kDa MWCO, (b) BSA-PESU 10 kDa MWCO, (c) lysozyme-Hydrosart 5 kDa MWCO, and (d) lysozyme-PESU 10 kDa MWCO.

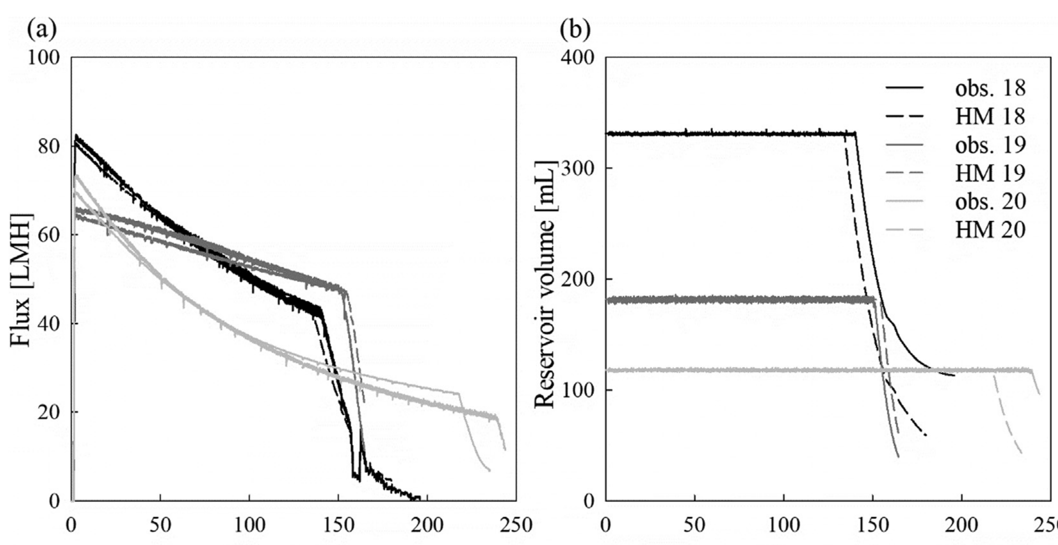

Fig. 6. Three fed-batch continuous UF and hybrid model predictions with varying TMPs, CFs, c_Bs, and starting reservoir volumes and a total bulk volume of approximately 3000 mL. (a) Flux vs. time and (b) reservoir volume vs. time.

therefore attract positive feed constituents strongly and has a substantial effect on the fouling of the membrane. In addition the difference in the size of lysozyme (14.7 kDa) and the pores of the membrane (10 kDa) is small and it is likely that pores are partially internally blocked by lysozyme adsorption in the pore channel [27]. These combined effects led to the quick fouling of the membrane and a drastic drop in flux compared to all other protein-membrane combinations.

3.5. Fed-batch continuous UF

The hybrid model built on training sets obtained by batch processes were also able to predict fed-batch processes. For concentration processes carried out in fed-batch mode, the white box was adjusted to account for the automated inflow from the feed bottle to the reservoir. Fig. 6 shows the three fed-batch processes for the concentration of BSA on Hydrosart membranes; here, the fed-batch volume was added according to the permeate flow to keep the reservoir level constant (Fig. 6b). As soon as the feed bottle was empty (inflection point in Fig. 6a), the regular batch process continued. The model assumed this change from fed-batch to batch at different times. For two out of three fed-batch processes (runs 18 and 19), there was very good agreement between observed and predicted values, while the third (run 20) was slightly overestimated (earlier switch from fed-batch to batch model) by the hybrid model (Fig. 6). The NRMSEs (Table 2) are similar to those of the batch processes, however, a slight misestimation of the inflection point between fed-batch and batch would have a large impact on the error. The fed-batch UF errors were all in a similar range, 2.6-5.9 NRMSE, which is impressive considering the longer process durations. Summarizing, the hybrid model was able to predict the flux of the fed-batch processes, even though the training was performed on data received from batch processes.

The prediction of fed-batch processes is only possible when using a mass balance–based white box. Pure black box models that use TMP, CF, and initial c_B as input and the filtration endpoint as output are not predictive for fed-batch processes, since they lack information such as reservoir level and c_B changes due to the incoming fresh sample. In the presented hybrid model, this information is supplied by the white box. Impressively, the model is also able to accurately account for the flux change due to temporarily reduced TMP by the operator (Fig. 6, run 18 at 160 min). Here, the hybrid model accurately predicted the resulting flux after changing the TMP after a previous fed-batch process that was not used in the model training.

3.6. Simulation study

As there was excellent hybrid model performance on the test data, as well as on the fed-batch processes, the model is able to very precisely describe the complete process and the overall filtration duration for a given CF, TMP, and initial c_B, especially for Hydrosart membranes (Table 2). Not only can the presented hybrid model predict the flux for a given set of input parameters, but it can also be used to calculate the required TMP for any desired duration for a given set of CF and initial c_B. The algorithm that was developed started at a TMP of 1 bar and was adjusted using Eq. (6). By subsequently adjusting TMP_{min} and TMP_{max} to optimize the $TMP_{desired}$ for each iteration, the desired TMP was reached after 5–10 iterations. Fig. 7 provides an example of how the flux and retentate volume changed throughout each iteration when using 250 mL/min CF, 4 g/L initial c_B of BSA with the Hydrosart membrane, and 80 mL retentate volume as the stop criterion. The boundary criterion for this method was that the TMP must not exceed the maximum TMP of the membrane (4 bar). After only six iterations, the desired TMP of 2.08 bar was reached (Fig. 7b). Alternatively, the initial c_B and CF can be estimated by the hybrid model for both the batch and fed-batch modes and therefore be used as a digital twin for process optimization.

3.7. Comparison to film theory

The flux predictions made by the hybrid model were compared to the predictions of the well-established film theory model. While the black box part of the hybrid model used c_B, CF, and TMP to predict the flux, the film theory model in Eq. (7) uses c_B, c_G, and k. The flux at a certain TMP and CF was plotted against the decadal logarithm of c_B to calculate c_G and k. The film theory assumes a logarithmic relationship between the flux and c_B, which arises when using a hydrophobic PESU membrane (Fig. 8a). The experiments with the Hydrosart membrane, however, deviated from this relationship when c_B was lower than 25 g/L. It appeared that, at low c_B, the flux already reached the pressure-independent region for the PESU membrane, but it was still pressure dependent for the Hydrosart. Therefore, film theory can only be used to model the flux of the PESU membrane. Since the film theory overestimated the fluxes, especially at low c_B, it cannot be used to describe the test set experiments over time to predict the correct process duration for Hydrosart UF (Fig. 8b). PESU experiments were predicted more accurately, because the flux was modeled well at low c_Bs (Fig. 8c). This

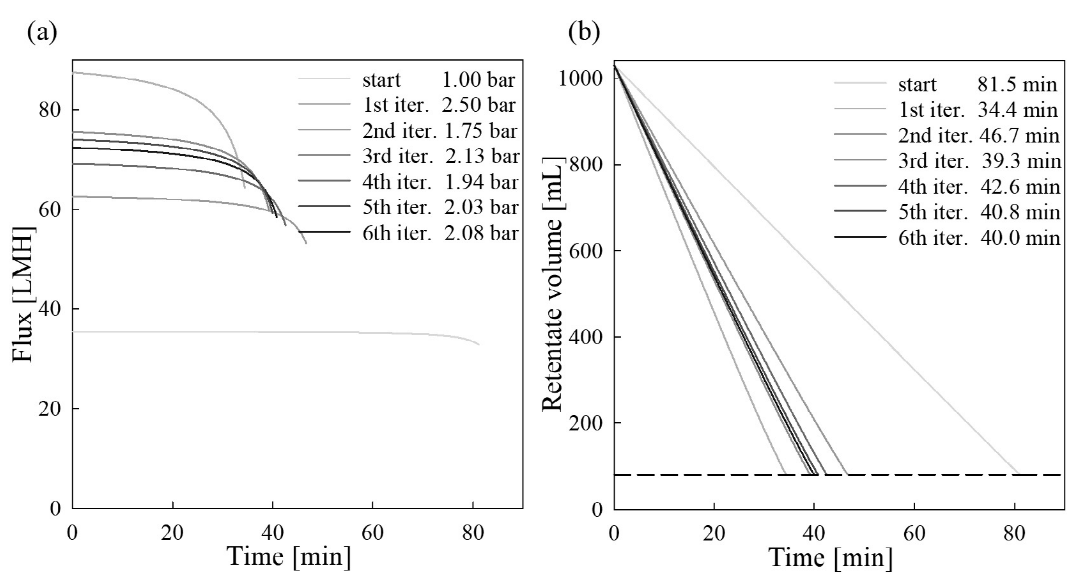

Fig. 7. Example of the automated TMP iteration to reach a retentate volume of 80 mL (dotted line) after 40 min with CF 250 mL/min and initial c_B of 4 g/L of BSA using Hydrosart. (a) Flux vs. time and (b) retentate volume vs. time.

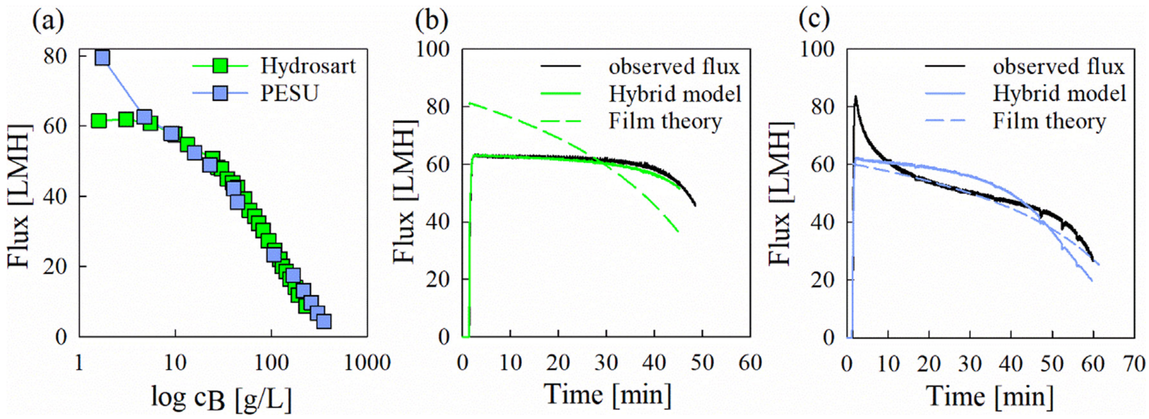

Fig. 8. BSA fluxes with two membranes. (a) Training set: flux vs. log(c_B) at CF 100 mL/min and TMP 1.8 bar. (b) Test set: flux vs. time for BSA-Hydrosart at CF 200 mL/min and TMP 1.8 bar. (c) Test set: flux vs. time for BSA-PESU at CF 200 mL/min and TMP 1.8 bar.

comparison was only possible when the chosen CF and TMP in the test set were identical to those in the training set.

When film theory was employed to predict a UF process, the desired parameters c_G and k had to be set via TMP and CF. For that purpose, good correlation was needed between k and c_G as well as TMP and CF. Fig. 9a,d shows a clear trend between k and TMP as well as between c_G and CF for BSA on the PESU membrane. High TMPs led to an increase in the convective flow toward the membrane; therefore, k increased. While c_G increased with higher CF (Fig. 9d), the c_G showed no correlation with TMP (Fig. 9c), demonstrating that it is pressure independent for a given protein-membrane combination [19]. Fig. 9b shows that k slightly decreased with increasing CF at TMP higher than 1.8 bar. A

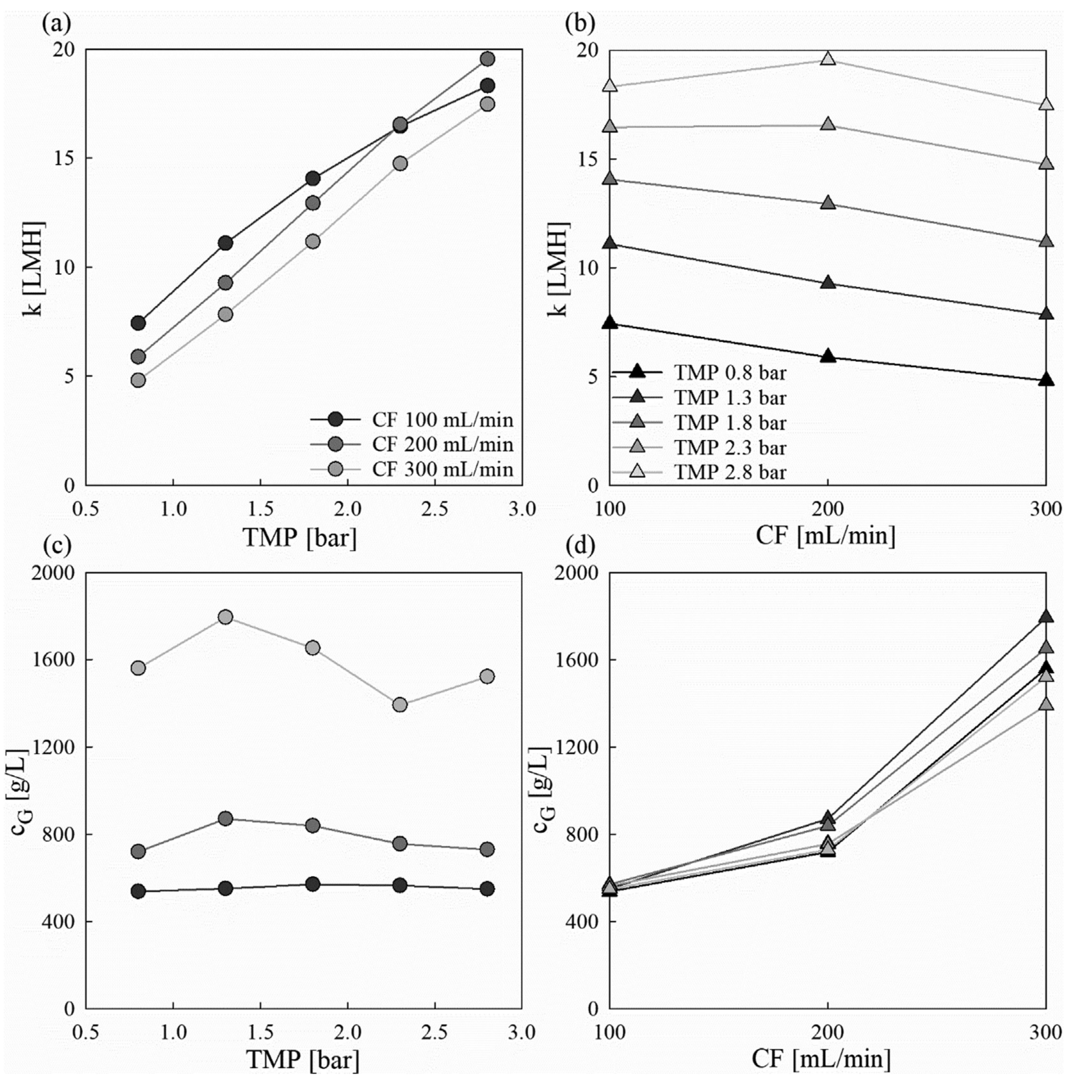

Fig. 9. Summary of the correlations between (a) k and TMP, (b) k and CF, (c) c_G and TMP, and (d) c_G and CF of BSA and the PESU membrane.

high CF reduced the convective flow toward the membrane, since the proteins were swept away faster. The high CF led to an increase in c_G, as shown in Fig. 9d, due to the higher concentration at the membrane surface leading to a shift in the mass transfer equilibrium.

We can compare film theory only to the black box models if the TMP and CF in the test run were identical to those for the training set. If film theory were to be applied to TMPs and CFs that were not covered in the training set, then the input parameters k and c_G would have to be estimated via interpolation, which would require a training set with an increased number of observations.

In contrast, the hybrid model can interpolate very well between TMP, CF, and c_B due to the high adaptability of the ANN and could be trained on comparably small training sets, which again highlights the potential of this application.

3.8. Training set reduction

The established methodology was based on several independent experiments to obtain the training data set consisting of up to three predictors. To reduce the workload in the lab and make the approach more efficient, we used reduced data sets from the original data without reducing the design space. The impact of a reduced data set on the predictive ability of the established model was then evaluated using the minimal NRMSE as the criterion. Various combinations of TMP, CF, and c_B were removed from the original data set in an equidistant manner, e.g., removing every other c_B reduces the lab work by 50%.

Using the entire training data set obtained for the concentration of BSA on the Hydrosart membrane, an NRMSE of 5.82 was obtained. After removing up to 50% of c_B data, with TMPs of 1.3 and 2.3 bar and a CF of 200 mL/min, the error remained similar at 4.30 NRMSE. However, if the data obtained for experiments at a TMP of 1.8 bar were also to be removed, then the NRMSE would increase 3-fold. The flux could therefore not be predicted reliably when only two TMP (0.8 and 2.8 bar) conditions were present in the training set. When the TMP was increased, the UF reached a pressure-independent region, resulting in a shallower slope (Fig. A.7b). This curve flattening could only be determined with at least three measurements. On the other hand, the CF was accurately predicted from only two CF conditions in the training set (100 and 300 mL/min), indicating a linear flux-CF relationship (Fig. A.7a), which was already reported for low CF [28]. Considering this for the data reduction, the time required for the training data generation could be reduced by 80% while still retaining a low NRMSE. The resulting NRMSEs for the black box part are summarized in Table A.2. A graphical illustration of the training set reduction is given in Fig. A.6. Hence, we showed that a drastically reduced training data set can still achieve high flux predictability over the entire process duration.

4. Conclusions

Process understanding and control are key to facing the challenges of increased demand for biopharmaceuticals. Monitoring and prediction of process deviations and corresponding control strategies are crucial for efficient and safe biopharmaceutical production. Predictive models form the basis of such approaches needed to implement Quality-by-Design concepts for manufacturing. Often, these mainly mechanistic models do not find their way to industrial applications due to their complexity and the modelers' lack of understanding.

We developed a simple bottom-up approach using hybrid models to predict the duration of cross-flow UF processes. Due to its excellent interpolation properties, the model can predict flux evolution and the duration of various combinations of input parameters within the training region with only a few experiments. Further, the hybrid model can predict the flux of both batch and fed-batch modes for concentration processes by using a generic white box, thereby expanding the predictions to large-scale operations. Since the black box remains untouched, it is not necessary to create a new training set because the flux prediction is c_B dependent (not time dependent).

While the pure mechanistic model already had difficulties with varying parameters because of limited interpolation capabilities, a pure black box model is not predictive for fed-batch processes. The suggested hybrid model is superior to both scenarios.

Considering the discussed advantages, the multistep-ahead hybrid model structure can be used as a digital twin to predict virtual processes with varying input parameters and different batch modes. The same model can be used for model predictive control, to adapt process parameters according to changes in the sample composition e.g. at higher product concentration TMP could be automatically increased to keep the process duration constant. Hence, the presented hybrid model structure represents a valuable tool for efficient process development, optimization and future control applications.

CRediT authorship contribution statement

Maximilian Krippl: Investigation, Methodology, Data curation, Formal analysis, Software, Visualization, Writing - original draft. **Astrid Dürauer:** Resources, Supervision, Writing - review & editing. **Mark Duerkop:** Conceptualization, Project administration, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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